

Summation (Sigma) Notation

Pre-Calculus / Algebra Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Read and interpret summation notation including identifying the function, lower limit, and upper limit
- Evaluate summation expressions by substituting each integer value from the lower to upper limit and summing the results
- Apply summation notation to a variety of functions including linear, quadratic, and rational expressions

Problems

1. Evaluate the summation of $3n$ from $n = 1$ to 4.

$$\sum_{n=1}^4 3n$$

2. Evaluate the summation of 5 from $n = 1$ to 4. (The function is the constant 5.)

$$\sum_{n=1}^4 5$$

3. Evaluate the summation of $(n + 4)$ from $n = 2$ to 5.

$$\sum_{n=2}^5 (n + 4)$$

4. Evaluate the summation of $(2n + 1)$ from $n = 0$ to 4.

$$\sum_{n=0}^4 (2n + 1)$$

5. Evaluate the summation of n squared from $n = 0$ to 3.



$$\sum_{n=0}^3 n^2$$

6. Evaluate the summation of $(x + 3)$ from $x = 4$ to 7.

$$\sum_{x=4}^7 (x + 3)$$

7. Evaluate the summation of $(3n \text{ minus } 2)$ from $n = 1$ to 5.

$$\sum_{n=1}^5 (3n - 2)$$

8. Evaluate the summation of $(n \text{ squared plus } 2n)$ from $n = 1$ to 4.

$$\sum_{n=1}^4 (n^2 + 2n)$$

9. Evaluate the summation of 2 divided by h from $h = 1$ to 4.

$$\sum_{h=1}^4 \frac{2}{h}$$

10. Evaluate the summation of $(n \text{ cubed minus } n)$ from $n = 2$ to 5.

$$\sum_{n=2}^5 (n^3 - n)$$



Summation (Sigma) Notation — Answer Key

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Answer Key

1. Answer: 30

- Substitute $n = 1, 2, 3, 4$ into $3n$: $3(1) + 3(2) + 3(3) + 3(4)$
 - Compute each term: $3 + 6 + 9 + 12 = 30$
-

2. Answer: 20

- The constant 5 is added once for each integer from 1 to 4, giving four terms: $5 + 5 + 5 + 5$
 - Sum = $4 \times 5 = 20$
-

3. Answer: 30

- Substitute $n = 2, 3, 4, 5$: $(2+4) + (3+4) + (4+4) + (5+4)$
 - Compute: $6 + 7 + 8 + 9 = 30$
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4. Answer: 25

- Substitute $n = 0, 1, 2, 3, 4$: $(0+1) + (2+1) + (4+1) + (6+1) + (8+1)$
 - Compute: $1 + 3 + 5 + 7 + 9 = 25$
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5. Answer: 14

- Substitute $n = 0, 1, 2, 3$: $0^2 + 1^2 + 2^2 + 3^2$
 - Compute: $0 + 1 + 4 + 9 = 14$
-

6. Answer: 46

- Substitute $x = 4, 5, 6, 7$: $(4+3) + (5+3) + (6+3) + (7+3)$
 - Compute: $7 + 8 + 9 + 10 = 34$... Re-check: $7+8=15$, $15+9=24$, $24+10=34$. Wait, let me recount. $7+8+9+10 = 34$.
Correction: answer is 34.
 - Sum = 34
-

7. Answer: 35

- Substitute $n = 1, 2, 3, 4, 5$: $(3-2) + (6-2) + (9-2) + (12-2) + (15-2)$
 - Compute: $1 + 4 + 7 + 10 + 13 = 35$
-

8. Answer: 50

- Substitute $n = 1, 2, 3, 4$: $(1+2) + (4+4) + (9+6) + (16+8)$
 - Compute: $3 + 8 + 15 + 24 = 50$
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9. Answer: 25/6

- Substitute $h = 1, 2, 3, 4$: $2/1 + 2/2 + 2/3 + 2/4$
 - Simplify each term: $2 + 1 + 2/3 + 1/2$
 - Find a common denominator of 6: $12/6 + 6/6 + 4/6 + 3/6 = 25/6$
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- Substitute $n = 2, 3, 4, 5$: $(8-2) + (27-3) + (64-4) + (125-5)$
 - Compute each term: $6 + 24 + 60 + 120$
 - Sum: $6 + 24 = 30$, $30 + 60 = 90$, $90 + 120 = 210$. Recheck: $8-2=6$, $27-3=24$, $64-4=60$, $125-5=120$. $6+24+60+120 = 210$
 - Sum = 210
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